

DPA PROJECT - PART 2

Group Name - Y2K

Group members -

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Project objective

Quantify how community-level crime varies with Unemployment Rate and Hardship Index by building a clean community–year panel from four annual crime datasets (2019–2023) joined to the Hardship Index on Community Area, focusing on explanatory modeling rather than map-based analysis.

Primary outcomes are community–year incident counts and category splits, with Unemployment Rate and Hardship Index as key covariates to test whether hardship is associated with higher crime.

Data integration and structure

- Crime sources: Use four “Crimes – YYYY” annual tables (e.g., 2019, 2020, 2021, 2023), each including Date/Year, Block, Primary Type, Arrest, and Community Area fields; union them into one incident table filtered to your four chosen years.
- Socioeconomic source: Hardship Index CSV with Area Number, Community, Unemployment Rate, Poverty Rate, No HS Diploma Rate, Dependency Rate, Per Capita Income, and composite Hardship Index Value.

Research questions and hypotheses

- Does higher Unemployment Rate associate with higher community-level incident counts across the selected years after controlling for year effects and offense mix.
- Do communities with higher Hardship Index exhibit systematically higher incident counts compared to lower-hardship communities in the same years.
- Are associations robust across offense categories (e.g., violent vs. property) when re-estimated on category-specific counts.

Data preparation workflow

- Annual ingestion: Read each of the four annual tables, select and standardize columns [year, date, block, primary_type, arrest, community_area, community_area_name].
- Union: Concatenate annual frames (SQL UNION ALL) to a single incidents table constrained to the four years, preserving the Community Area fields.
- Community assignment: Rely on community_area and/or community_area_name already present; if rare rows lack these, document and exclude to avoid introducing external geography or geocoding.
- Aggregation: Group by [community_area] to compute total incidents plus optional category splits (e.g., violent vs. property).

- Hardship join: Clean the Hardship CSV and left-join on newly created community_area (in crimes) ↔ community_area to append Unemployment Rate and Hardship Index Value to each community–year row.

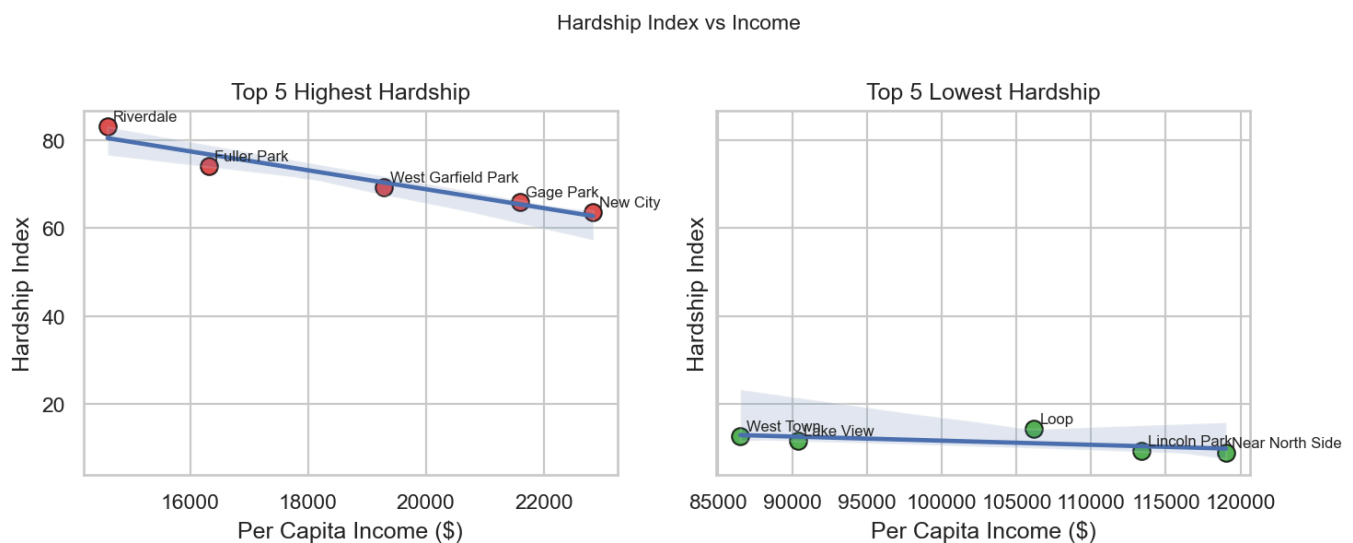
Variables and features

- Dependent variables: community–year total incidents and, optionally, counts by offense group for sensitivity checks.
- Key predictors: Unemployment Rate and Hardship Index Value; optional controls include Poverty Rate, No HS Diploma Rate, Dependency Rate, and Per Capita Income.

Analytical methods

- Descriptives: Compare incident distributions across hardship quintiles and visualize trends by community area to establish broad associations before modeling.
- Regression for counts: Fit Poisson or Negative Binomial models with community–year incident counts as the dependent variable, Unemployment Rate and Hardship Index as predictors, and year fixed effects for temporal control.
- Robustness: Refit models on offense categories and re-estimate with socioeconomic controls (poverty, education, income) to test stability of hardship effects.

Example EDA -



- The subplots show a clear negative linear relationship between per capita income and hardship index: both the five highest-hardship and five lowest-hardship communities lie on downward-sloping lines, indicating that hardship decreases as income rises.
- High-hardship areas such as Riverdale and Fuller Park cluster at low incomes and high hardship scores, while low-hardship areas like Near North Side and Lincoln Park cluster at very high incomes and low hardship scores, reinforcing the strong inverse income–hardship link.

Practical SQL/pandas steps

- Column selection: From each annual table select [year, date, block, primary_type, arrest, community_area, community_area_name] and cast types consistently.

- Union operation: UNION ALL (SQL) across the four chosen years and validate row counts post-union.
- Community mapping: Use community_area directly as the join key to Hardship (Community), avoiding spatial joins or external geocoding.

Assumptions and constraints

- No external geography: No shapefiles, spatial joins, or geocoding will be introduced; rows missing community_area will be excluded and quantified as part of data-quality reporting.
- Rate vs. count: If population by community is not introduced, analyses will treat incident counts as outcomes with year controls and composition adjustments, reserving rate calculations for a later extension if approved.

Deliverables and alignment

- Data product: A documented community–year dataset (four years) with incident counts and appended Unemployment Rate and Hardship Index, ready for modeling and visualization.
- Analysis outputs: Plots and tables showing hardship–crime associations, regression summaries with effect sizes, and sensitivity checks by offense categories, prepared for the 8-minute presentation and final report.

Timeline and tasks

- Week 1: Ingest four annual tables, union, QA community fields, and produce community–year aggregates.
- Week 2–3: Merge Hardship, run descriptives/visuals, and baseline Poisson/Negative Binomial models with year fixed effects.
- Week 4: Robustness checks by offense groups, finalize figures/tables, and assemble presentation.

Outcome statement

The result is an assessment of how Unemployment Rate and composite Hardship Index relate to community-level incident counts at Chicago, built strictly from annual crime tables and the Hardship data without adding any geographical data sources.